

Smart Energy Meter - User Guide

How a customer installs and uses the mobile app and smart plug.

1. Install the Mobile App

- 1 Download the APK file named SmartEnergyMeter.apk from the shared link.
- 2 Tap the APK file on the Android phone.
- 3 If Android asks permission, enable Install unknown apps for the file manager/browser.
- 4 Tap Install.
- 5 Open the Smart Energy Meter app.

2. First Wi-Fi Setup

- 1 Power ON the smart plug.
- 2 Press and hold the Wi-Fi setup button on the device.
- 3 The OLED will show the setup Wi-Fi name such as SmartPlug_XXXX.
- 4 On the phone, open Wi-Fi settings and connect to SmartPlug_XXXX.
- 5 Open the app and go to Settings -> ESP32 Wi-Fi Setup.
- 6 Enter the home Wi-Fi name and password.
- 7 Tap Send Wi-Fi to ESP32.
- 8 The device restarts and connects to the home Wi-Fi.

3. Daily Use

- Open the app to view live voltage, current, power, energy, and cost.
- Use the relay switch in the app to turn the plug ON or OFF.
- Use the timer to automatically turn OFF the plug after a selected time.
- Use history and analysis pages to check energy usage.

4. Safety Notes

- This project works near AC mains voltage. Build and test it only with qualified supervision.
- Use proper enclosure, insulation, strain relief, fuse/MCB protection, and safe earthing where required.
- Do not touch any AC/live part while powered.
- This documentation shows system-level connections. Final mains wiring must be checked by a qualified person before use.